
ICPC Algorithm Library

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目录

1	Basic	1
1.1	test	1
2	Geometry	1
2.1	点和向量	1
2.2	线段和多边形	1

1 Basic

1.1 test

```
export fsan="-fsanitize=address,undefined"
ulimit -s 1024000
ulimit -v 1024000
mk() { g++ -o $1 $1.cpp -O2; }
```

```
pai() {
    mk a && mk bf && mk gen || return
    cnt=0
    while true; do
        ./gen > 1.in
        ./bf < 1.in > bf.out
        ./a < 1.in > a.out
        if diff -Zq a.out bf.out; then
            echo "$((++cnt)) AC"
        else
            echo WA
            break
        fi
    done
}
```

2 Geometry

2.1 点和向量

```
using db = long double;
struct p2 {
    db x, y;
    db len() { return hypot(x, y); }
    p2 operator+(p2 b) { return {x + b.x, y + b.y}; }
    p2 operator-(p2 b) { return {x - b.x, y - b.y}; }
    p2 operator*(db k) { return {x * k, y * k}; }
    p2 operator/(db k) { return {x / k, y / k}; }
    db operator*(p2 b) { return x * b.x + y * b.y; } // 点积
    db operator%(p2 b) { return x * b.y - y * b.x; } // 叉积
};
using ps = vector<p2>;
```

2.2 线段和多边形

```
// 点到直线距离
db toL(p2 p, p2 a, p2 b) { return abs((b - a) % (p - a)) / (b - a).len(); }
```